

Violent Crime Rate in California

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1 Introduction and Goals

This project analyzes the *Violent Crime Rate in California* dataset from 2000 to 2013, sourced from U.S. Open Data. The dataset offers a valuable opportunity to explore how violent crimes have evolved across California's counties, and how these patterns relate to broader socio-economic conditions.

To enrich our analysis, we manually incorporated additional data on county-level GDP, enabling us to study the potential influence of economic development on crime rates. This augmentation allowed for a more comprehensive examination of regional disparities.

Our objectives were fourfold. Firstly, we aimed to visualize the geographic distribution of violent crimes across California. Our goal was to identify spatial clusters and temporal trends. Secondly, we applied Principal Component Analysis (PCA) to uncover latent structures and correlations among variables. Thirdly, we evaluated the predictive power of different feature combinations using a k-Nearest Neighbors (KNN) classifier. We did so by comparing the impact of crime types with socio-economic attributes. Lastly, we developed a recommender system based on cosine similarity to identify counties with similar crime and economic profiles, facilitating insights into regional policy transfer and collaborative public safety planning.

This multidimensional approach enables a deeper understanding of violent crime dynamics across the state and demonstrates the value of integrated data analysis in social research.

2 Understanding the Data Contents and Structure

The original dataset, sourced from U.S. Open Data, consisted of over 49,000 entries detailing the crimes that occurred during the 2000 to 2013 time period. However, the structure and granularity of the data required refinement to suit our analytical objectives.

Each record in the dataset represented a single felony type for a given geographic area and year, resulting in multiple entries for the same location. To streamline our analysis, we aggregated the data by city and year, condensing repeated entries into a unified format that preserved historical depth while reducing redundancy.

We also identified the attributes that were either irrelevant or inconsistent with our research goals and excluded them from the working dataset. The remaining ones allowed us to maintain critical geographic identifiers (e.g., county name, FIPS codes, region) as well as essential variables such as population estimates and crime counts.

Furthermore, the original dataset provided only raw counts for crimes and population, without calculating standardized crime rates. To enhance the comparative analysis across counties, we computed the violent crime rate per 1,000 inhabitants directly from the provided values. This transformation permitted us to incorporate both the scale and intensity of violent crime into subsequent analyses.

2.1 Selected Attributes

The attributes we considered essential are the listed following:

- **geoname**: The name of the geographic area.
- **geotypevalue**: A numerical code associated with the geographic type.
- **county_name**: Name of the county.
- **county_fips**: Federal Information Processing Standard (FIPS) code for the county.
- **region_name**: The name of the region.
- **region_code**: A numerical code of the region.
- **dof_population**: The Department of Finance population estimate.
- **AA**: Division for aggravated assault crimes committed.
- **FR**: Division for forcible rape crimes committed.
- **M_and_NNM**: Division for murder and nonnegligent manslaughter crimes committed.
- **R**: Division for robbery crimes committed.
- **VCT**: Division for total violent crimes committed.
- **rate**: The number of violent crimes per 1000 population.
- **gdp_dollars**: GDP data of 2023 for further computations.
- **lat**: Latitude coordinate.
- **lon**: Longitude coordinate.

(The last three attributes were added by us)

3 Geographic Visualization

A central objective of our analysis was to visualize the spatial distribution of violent crimes across California. By mapping incidents geographically, we aimed to identify regional patterns and observe how these evolved over time.

3.1 Geocoding Coordinates

The original dataset did not include geographic coordinates. To resolve this, we employed the `geopy` library and used the Nominatim API from OpenStreetMap to geocode the `geoname` entries. This process converted location names into latitude and longitude values suitable for mapping. During geocoding, a small number of entries failed due to naming inconsistencies, such as the inclusion of terms like “city” or nonstandard formats. These were reviewed and corrected manually, ensuring the integrity of the spatial data.

3.2 Plotting Process

After enriching the dataset with coordinates, we merged this information with the cleaned dataset and used `matplotlib` to produce geographic visualizations. We overlaid crime data points on a background image of the state of California to ensure accurate geographic placement.

We created individual maps for each year since the 2000 to 2013, allowing us to visually track shifts in crime intensity and distribution over time. These plots helped reveal persistent crime hotspots plus regional crime dynamics that would not have been as evident from tabular analysis alone.

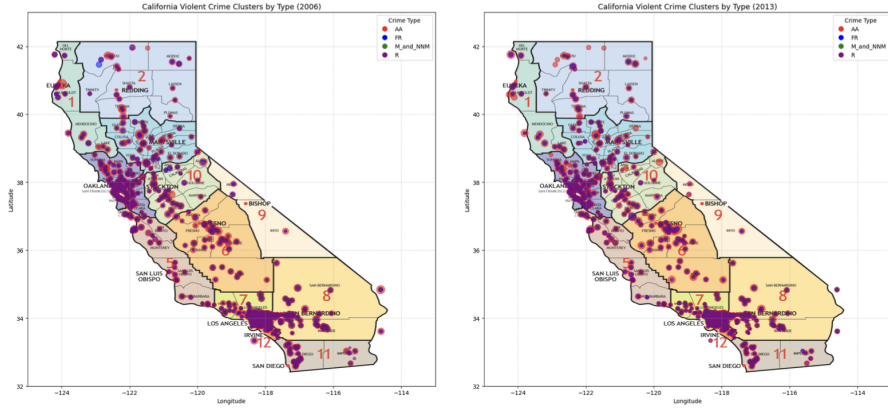


Figure 1: Sample of the plots (2006 and 2013)

4 Exploratory Data Analysis

In this section, we analyzed the data to prepare it for machine learning, by exploring feature distributions, potential cluster structures, class separability, and dimensionality. This step was crucial for identifying potential outliers, assessing feature distributions, checking for class imbalance, and informing subsequent pre-processing decisions.

4.1 Class Separability and Clustering Potential

To assess whether meaningful groupings existed in the data, we used scatter plots and pairwise feature comparisons. These visualizations helped distinguish potential clusters based on crime characteristics and socio-economic indicators.

In addition, a pair plot of `rate`, `dof_population`, and `reportyear`, as seen in Figure 2, provided a multivariate view of how these features related. It revealed a wide spread in both crime rate and population, supporting the idea that clustering may differentiate counties by crime intensity and scale. While no clear clusters were immediately visible in two-dimensional space, the presence of structure in the data justified the use of dimensionality reduction to further explore latent groupings.

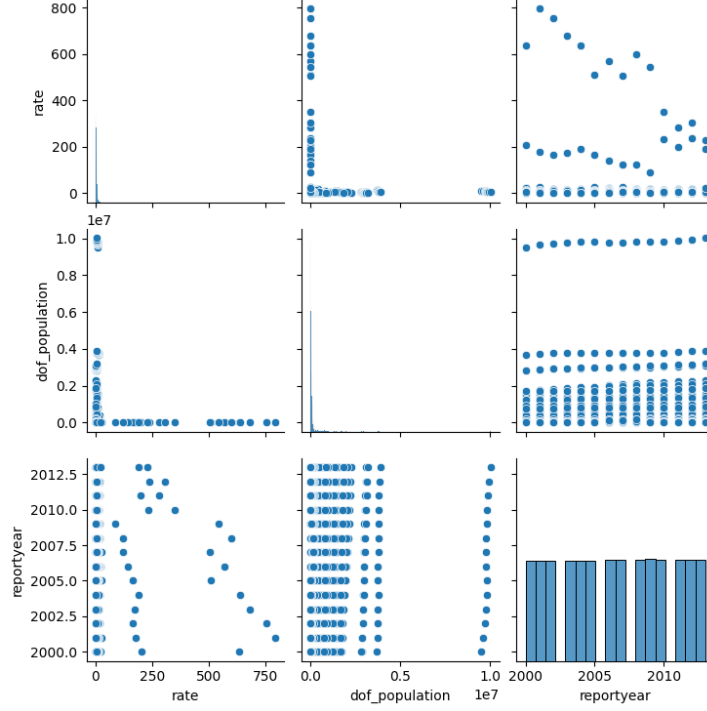


Figure 2: Feature Relationships for Clustering

4.2 Dimensionality Considerations

During our analysis, we observed high correlations and potential redundancy among several crime-related features, such as AA, VCT, and R.

To advocate this, we used pair plots to examine relationships between multiple features simultaneously. This multivariate view, shown in Figure 3, reinforced our findings and suggested that the dataset contained overlapping information that could be compressed without significant loss.

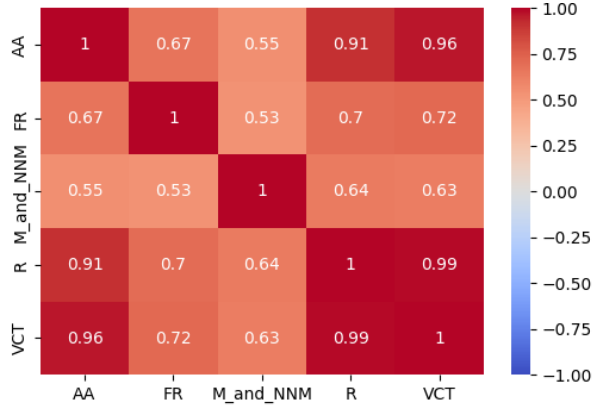


Figure 3: Correlation Between Crime Types

As a result, dimensionality reduction appeared to be a promising strategy for simplifying the feature space while retaining its informative structure. This insight motivated the use of Principal Component Analysis (PCA), which we will explore in the following section.

4.3 Principal Component Analysis

We have applied PCA on all numerical features in the dataset, which includes variables such as **rate**, **dof_population** (population estimate), **gdp_dollars** (gross domestic product in dollars) and potentially other numeric fields in the dataset.

Before applying PCA, the data was cleaned, by removing rows with missing values, and standardized using `StandardScaler` to ensure that all variables contributed equally to the analysis, regardless of their original scale. The PCA output enlightened us the following:

- The first component explains 38.06% of the variance in the data.
- The second component, when added, explains a cumulative 57.87% of the variance.
- By the 11th component, nearly 100% of the variance is explained, and after the 12th component, the cumulative variance stops increasing, remaining at 1, indicating that all the variance is captured.

This suggests that the data can be effectively reduced to a smaller number of components without losing much information. For example, the first 5 components could be retained to explain around 82.26% of the variance.

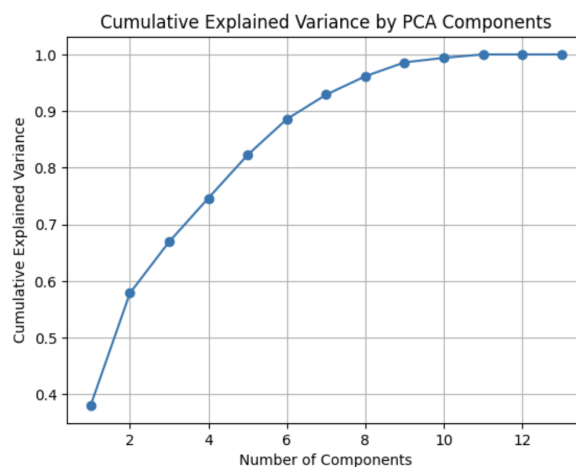


Figure 4: Plot of the cumulative explained variance by PCA components

PCA has helped us recognize underlying patterns and correlations between variables such as crime rate, population, and GDP in California counties and years.

5 K-Nearest Neighbors (KNN)

In this section of the study we developed our KNN model, first to filling in the missing gaps of our data and later, building it to test its performance. This is a quite important step in order to obtain the most accurate results.

That is why we started with encountering the missing values. As we can not move forward without having complete information if we want a proper performance for our model. So, we used the `.isnull().sum()` to do so. This way we found how many and in which features elements were missing.

Moving forward we choose imputation as our strategy to tackle down the inconvenience. Initially we wanted to use the median to fill in our gaps, but we wanted to make sure before proceeding. That is the reason behind calculating it along with the mean. Consequently we noticed the numeric difference between both values and reckoned we were in the right. As the median provides a more balanced imputation and is better when handling outliers.

```
mean_gdp = df["gdp_dollars"].mean()
median_gdp = df["gdp_dollars"].median()
difference = mean_gdp - median_gdp
print(f"Mean GDP: {mean_gdp:.2f}")
print(f"Median GDP: {median_gdp:.2f}")
print(f"Difference between the Mean GDP and Median GDP: {difference:.2f}")
```

✓ 0.0s Python

Mean GDP: 128489749.12
Median GDP: 46573826.00
Difference between the Mean GDP and Median GDP: 81915923.12

Figure 5: Output of mean and median

We picked rate as our target. It was the main variable of interest due to its numeric and continuous values plus its dependence to the other measured inputs (AA, FR, M_and_NNM, R, VCT). We also categorized the column into three equal groups (small, medium, high). Right after, we were positive to consider two different groups of features for our model. These being `county_fips` with `gdp_dollars` and the different types of crimes. All that was left was choosing the best option.

Moreover we used `imputer = SimpleImputer(strategy="median")` to replace the missing values in each of the features and `scaler = StandardScaler()` to standardize them.

Moving on we carried out the train-test split. To achieve it we applied `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25 random_state=42)` where we split our data 75% for our training set and the remaining 25% for our test set. This ensured we had enough data to train the model and for evaluating how well it manages new data. This evaluation is useful to prevent overfitting in our model.

In addition, we needed to see how our KNN would act with multiple k values. For that reason we defined a list these as test values: `k_test_values = [1, 3, 5, 7, 10]`. We reckoned these values would work the best as odd numbers prevent ties and it is a reasonable range for the comparisons. Further, we used `score = clf.score(X_test, y_test)` to compute the accuracy for every test value.

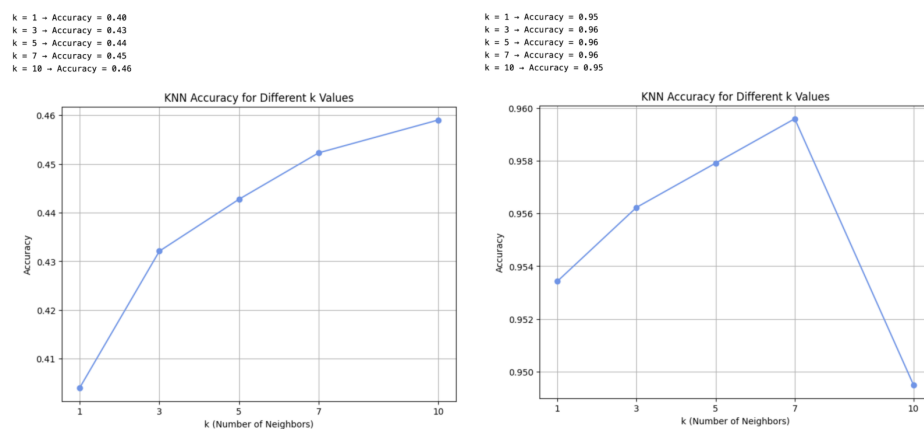


Figure 6: Accuracy vs k values for `gdp_dollars` and `county_fips` and different types of crimes

To finish up we decided to develop a plot to be able to visualize the k values in the range with their corresponding accuracy value. These were the results obtained:

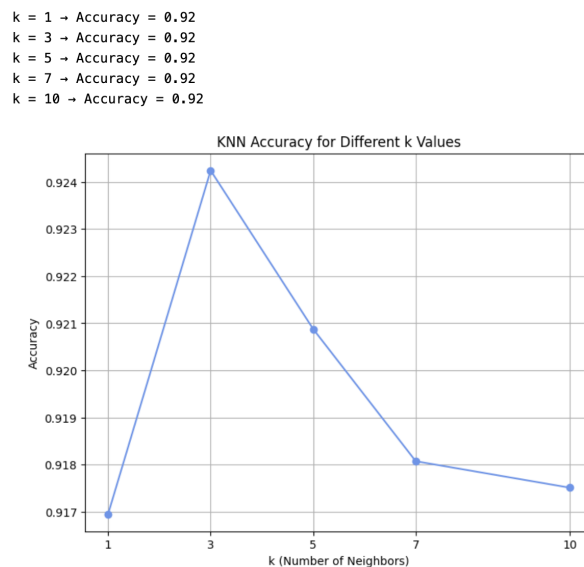


Figure 7: Accuracy vs k values for `gdp_dollars` and different types of crimes

We developed the same exact code for both using as a reference the steps done in the KNN Lab

assignment. That's when we noticed that the different types of crimes worked better as features. The reason behind is a higher accuracy for every one of the k test values, given that they are correlates of rate, the target variable. Still we wondered how would our results change if we added `gdp_dollars` as a sixth feature. We simply reused our code and run it. Despite adding it resulted in having a distinctively high accuracy for each values, they did not overcome the previously obtained results.

6 Recommender System

To explore structural similarities among counties in California with respect to crime and socio-economic indicators, we developed a recommender system based on cosine similarity. The primary goal was to find out which were the counties with similar violent crime profiles, potentially enabling comparative policy analysis or collaborative planning.

The following attributes were used to characterize each county:

- **dof_population**: Population estimate provided by the Department of Finance.
- **AA**: Count of aggravated assault crimes.
- **FR**: Count of forcible rape crimes.
- **M_and_NNM**: Count of murder and nonnegligent manslaughter crimes.
- **R**: Count of robberies.
- **rate**: Violent crimes per 1,000 inhabitants.
- **gdp_dollars**: Manually added GDP data at the county level to represent economic context.

These features captured both the absolute scale and proportional intensity of violent crimes, as well as the demographic and economic environment of each county.

6.1 Preprocessing

Before computing similarities, we removed rows with missing values in the selected attributes. All numerical features were then standardized using `StandardScaler` to ensure fair comparison by eliminating scale-related bias.

6.2 Recommendation Strategy

We computed the cosine similarity matrix using `sklearn.metrics.pairwise.cosine_similarity`, where each row and column represented a county. Cosine similarity measures the cosine of the angle between two feature vectors, effectively comparing the shape of their distributions rather than their magnitudes.

For each county, the system identified the top- k most similar counties. This facilitated our work when it came to detect structural parallels between counties that might not have been geographically close but shared similar crime and socio-economic characteristics.

This information was valuable for:

- Transferring successful intervention strategies to similar counties.
- Anticipating future trends by observing comparable counties.
- Designing region-specific public safety and policy measures.

6.3 Results and Analysis

The recommender system revealed multiple clusters of structurally similar counties. Urban areas with high population and GDP, such as Los Angeles and San Diego, often formed tight similarity groups, while rural counties tended to form separate clusters. These insights suggested shared challenges and the potential for coordinated crime prevention strategies within each group.

	County	Similarity
0	San Francisco	0.837684
1	Sacramento	0.791031
2	San Bernardino	0.763691
3	Alameda	0.714326
4	Monterey	0.572675
5	Kern	0.536228
6	San Joaquin	0.528420
7	Fresno	0.472290
8	Tulare	0.458841
9	Merced	0.419088

Figure 8: Sample output for Los Angeles county

While the current implementation effectively recognized structural similarities between counties, the recommender system could be further accentuated by incorporating temporal data or additional socio-economic indicators such as unemployment or education levels. These extensions would result in more dynamic and context-rich recommendations for policy and planning purposes.

7 Conclusions

This project provided a comprehensive analysis of violent crime data in California from 2000 to 2013, enhanced with additional socio-economic indicators to uncover structural patterns and regional disparities. By preprocessing and aggregating the original dataset, we enabled meaningful comparisons between counties while preserving essential historical and geographic information.

We applied dimensionality reduction through Principal Component Analysis (PCA) to identify latent structures in the data and evaluated different feature combinations using a K-Nearest Neighbors (KNN) classifier. These models helped assess the predictive value of crime types and socio-economic variables in classifying crime intensity.

A key contribution of this work was the development of a recommender system based on cosine similarity, which allowed us to identify counties with similar crime and economic profiles. This system opens the door for regional policy benchmarking and more informed public safety planning.

Overall, our results highlight the importance of integrating demographic, economic, and criminal data to better understand the dynamics of violence in California. The tools and methodologies we implemented can serve as a foundation for further research or public policy applications.

8 References and Annex

For any further information needed, you can check our github repository, with all datasets used, and other assets if useful: https://github.com/gyahiacheikh/VCRC_Data

Reference dataset link: <https://catalog.data.gov/dataset/violent-crime-rate-9a68e>